Time: 1 hr. Paper Code: CEN-504	B. Tech. (V Semester) 2nd Sessional Test 2024-25 Object Oriented Programming	Max. Marks: 15

Note: Attempt All Questions.
Write all answers using Java.
Suitable assumptions can be made wherever necessary.

Q1. CO4 (a) What is the output of the following code snippets? 2 Marks [5]

```
class MyTest{
  public static void main (String[] args)
  {
    int arr1[] = {1, 2, 3};
    int arr2[] = {1, 2, 3};
    if (arr1 == arr2)
      System.out.println("Same");
    else
      System.out.println("Not same");
  }
}
```

```
class X {
  static void method() {
    System.out.println("Static Method in X");
  }
}

class Y extends X {
  static void method() {
    System.out.println("Static Method in Y");
  }
}

public class TestDemo {
  public static void main(String[] args) {
    X obj = new Y();
    obj.method();
  }
}
```

(b) Explain briefly each keyword in the following Java method declaration. 2 marks

public static void main (String [] args)

(c) Differentiate between the following features of Java: Platform Independent and Architecture-neutral. 1 mark

2. O4 Define a class named *Employee* with *employee_id* and *employee_name* as its members. Extend this class by creating a subclass called *Salary*, which includes *designation* and *monthly_salary* as additional members. Implement the following: [5]

(a) Appropriate constructors for initializing the members.
(b) A method to find and display the details of employees who earn more than Rs. 20,000.
(c) A main method to create an array for storing employee details and demonstrate the functionality.

3. O5 Create a base class *Customer* with a method *getDiscount()* that provides a default discount (e.g., 0%). Then, implement three subclasses: *RegularCustomer* (10% discount), *PremiumCustomer* (20% discount), and *VIPCustomer* (30% discount). Use *runtime polymorphism* to dynamically determine and apply the appropriate discount based on the actual customer type, calculating the final price accordingly. [5]

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